

Program: Diploma in Food Processing Technology	
Course Code: 3425	Course Title: Food safety, quality, and hygiene
Semester: 3	Credits: 0
Course Category: Program core	
Periods per week: 3 (L: 3T:0 P: 0)	Periods per semester: 45

Course Objectives:

1. To understand and apply principles of Food Science to control and assure the quality of food products
2. To implement quality control in food processing industries
3. To impart knowledge on implementation of quality assurance systems

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food safety	Food safety, quality, and hygiene

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the principles that make a food product safe for consumption	12	Understanding
CO2	Identify the principles of Food Science to control and assure the quality of food products	12	Understanding
CO3	Review of quality control in food processing industry	10	Understanding
CO4	Describe the implementation of quality assurance system	11	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the principles that make a food product safe for consumption		
M1.01	Identify the different types of hazards	4	Understanding
M1.02	Describe design, construction, cleaning and disinfection of facilities	4	Understanding
M1.03	Explain different methods of pest control	4	Understanding
Contents:- Definition of hazards, types of hazards with examples, Design and construction of food premises, cleaning and disinfection facilities, Control of rats and rodents, insects and microbes, Personal hygiene			
CO2	Identify the principles of Food Science to control and assure the quality of food products		
M2.01	State the Principles of food safety	2	Understanding
M2.02	Illustrate the Contamination and cross contamination and its prevention	3	Understanding
M2.03	Discuss Food Recall and Traceability	3	Understanding
M2.04	Outline risk Assessment in food sector	3	Understanding
	Series Test –I	1	

Contents: Food Safety, Principles of food safety, Contamination and cross contamination, methods of preventing contamination, Food Recall and Traceability, Risk Assessment			
CO3	Review of quality control in food processing industry		
M3.01	List the factors affecting quality in the industry	4	Understanding
M3.02	Review the quality aspect in food supply chain	3	Understanding
M3.03	Summarize Quality control tools and evaluation methods	3	Understanding
Contents: factors affecting food quality, quality evaluation methods, Different quality control tools, Quality aspect in food supply chain and food processing industry			
CO4	Describe the implementation of quality assurance system		
M4.01	Describe HACCP	4	Understanding
M4.02	Explain ISO 22000	3	Understanding
M4.03	Describe Internal audit and documentation	3	Understanding
	Series Test–II	1	
Contents: HACCP, Principles of HACCP and steps in HACCP, CCP Determination, ISO 22000, Internal Auditing, Documentation and application of quality assurance in food industry			

Text/Reference

T/R	Book Title/Author
R1	Andres Vasconcellos J. 2005. Quality Assurance for the Food industry - A practical approach. CRC press.
R2	Inteaz Alli. 2004. Food quality assurance - Principles & practices. CRC Press. New York
R3	Sara Mortimore and Carol Wallace. 2013. HACCP - A practical approach. Third edition. Chapman and Hall, London.

R4	Roday, S. 1998. Food Hygiene and Sanitation, Tata McGraw-Hill Education.
R5	Neal D.Fortin. 2009. Food regulation, Wiley Publishers. FOOD TECHNOLOGY
R6	Naomi Rees. David Watson. 2000. International standards for food safety, An Aspen Publications.
R7	O'Rourke. 2005. European Food law, 3rd Edition, Thomson, Sweet and Maxwell.

Online Resources

Sl.No	WebsiteLink
1	https://www.techtarget.com/searchsoftwarequality/definition/quality-assurance
2	https://www.fda.gov/food/hazard-analysis-critical-control-point-haccp/haccp-principles-application-guidelines
3	https://www.highspeedtraining.co.uk/hub/what-is-the-food-supply-chain/